Environmental Monitoring

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Environmental Monitoring (A26962)

Short Title:	Environmental Monitoring
Faculty	Science and Computing
Department	Science
Cluster	Environmental Science
Author	MBREEN
Credits	5

Level: Introductory

Description of Module / Aims

This module will give students a broad introduction to the major elements of environmental science, from habitats, to the hydrological cycle, to global air pollution issues. A series of laboratory-based experiments plus field trips will be used to reinforce the theory elements of this module.

Programmes

`SCIE-0059`	BSc (Hons) in Food Science and Innovation (`WD_SFSIN_B`)	2 / 4 / M
`SCIE-0059`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	2 / 4 / M
`SCIE-0019`	BSc (Hons) in Pharmaceutical Science (`WD_SPHSC_B`)	1 / 2 / E
`SCIE-0059`	BSc in Food Science with Business (`WD_SFDSC_D`)	2 / 4 / M
`SCIE-0059`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	2 / 4 / M
`SCIE-0059`	BSc in Pharmaceutical Science (`WD_SPHSC_D`)	2 / 4 / M

Indicative Content

- Ecology: biodiversity, ecosystems, habitats, niche; species interactions competition, predation, parasitism, mutualism; succession, climax community; energy flow, food chains & webs
- Pollution: origins of pollution; pollutant monitoring; effects of pollution
- Water: hydrological cycle; pollution control, including wastewater and water treatment; EU Water Frameworks Directive
- Air: global air pollution issues, i.e. acid deposition, global warming, ozone depletion, international protocols
- Energy: clean energy technologies, energy efficiency/policy/poverty/security, electricity, natural gas, nuclear, oil, renewables/alternative energy, transport.
- Field trip to river ecosystems, including measurement of biotic index of water quality
- Laboratory practicals: physico-chemical analysis of water, i.e. hardness, BOD, nitrate and phosphate in water; microbiological analysis of water

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discuss how energy generation impacts the environment.
2. Apply ecological concepts (such as biodiversity indices, symbiosis) to the monitoring of ecosystems.
3. Describe the Greenhouse Effect and outline the scientific evidence behind anthropogenic climate change.
4. Discuss how pollutants are produced and monitored in the environment.
5. Explain the main topics associated with water in our environment, such as the hydrological cycle, pollution, treatment.
6. Identify significant air pollution components and explain their significance in terms of acidity and ozone depletion.
7. Identify the major ecological and physico chemical parameters necessary to access environmental quality.
8. Classify environments based on laboratory and field investigations.

Learning and Teaching Methods

- Lectures on the indicative content will be delivered, incorporating active learning activities.
- In-class and independent study assignments will be used to continually assess students' learning.
- Laboratory investigations plus field trips will reinforce and support theoretical elements of the module.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	4,7,8
In-Class Assessment	25%	1,3,6
Assignment	25%	2,5

Assessment Criteria

<40%: No understanding of the basic principles of environmental science.
40%-49%: Is able to understand the principles of environmental science. May have some difficulty with applying this theory.
50%-59%: All the above in addition to evaluation of practical data to assess environmental quality.
60%-69%: In addition to the above plus show a high level of competence and efficiency in the subject area.
70%-100%: All previous to excellent level. Be able to describe and discuss environmental science.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "Irish EPA website." www.epa.ie
- "Module Moodle page." http://moodle.wit.ie
- "The EU Environment ." www.eea.eu.int

Supplementary Material

- "International Panel on Climate Change ." www.ipcc.ch
- "US EPA website." www.epa.com
- Baird, C. and M. Cann. _Environmental Chemistry_. 5th ed. New York: W.H. Freeman, 2012.
- Lehane, M. and B. O'Leary. _Ireland's Environment - An Assessment_. Wexford: Environmental Protection Agency, 2012.
- Saksena, D.N. and D.M. Gaidhane. _Environmental Biology_. New Delhi, India: Studium Press, 2010.
- Spiro, T. _Chemistry of the Environment_. 3rd ed. United States: University Science Books, 2012.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Chemistry Lab